Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
ESC101	Engineering Mechanics	02	-	-	02	-	-	02

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment Test (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
ESC101	Engineering Mechanics	20	20	40	60	02	--	--	100

Rationale:

Engineering mechanics is a branch of science that deals with the behavior of solid bodies when subjected to external forces or loads and the effects of these forces on the bodies. It is a fundamental discipline within engineering and provides the basis for understanding and analyzing various types of structures and mechanisms.

Course Objectives:

1. To acquaint with basic principles of centroid and its application
2. To familiarize with the concepts of force, moment, Resultant and Equilibrium of system of coplanar force.
3. To acquaint with the basic concept of friction and its application in real-life problems.
4. To understand the parameters required to quantify the Kinematics of Particle and Rigid body.
5. To understand the parameters required to quantify the Kinetics of rigid body.
6. To acquaint with the basics of Robot kinematics

Course Outcomes:

1. Determine the equivalent force-couple system for a given system of forces.(L3)
2. Demonstrate the understanding of Centroid and its significance and locate the same. (L3)
3. Illustrate the concept of force, moment and apply the same along with the concept of equilibrium in two- and three-dimensional systems with the help of FBD. (L3)
4. Calculate position, velocity and acceleration etc. of particle/rigid body using principles of kinematics (L3)
5. Analyze particles in motion using force and acceleration, work-energy and impulse-momentum principles (L4)

6. Establish the relation between robot joints and parameters (L2)

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Resolution of a force. Use of trigonometry functions. Parallelogram law of forces. Law of triangle. Polygon law of forces, Lami's theorem. Concepts of Vector Algebra. Uniformly accelerated motion along a straight line, motion under gravity, projectile motion, Time of flight, Horizontal range, Maximum height of a projectile. Law of conservation of Energy, Law of conservation of Momentum, and Collision of Elastic Bodies. Work-Energy Principle (Note: There will be no questions from the prerequisite in the theory examination)	01	CO1
I	**System of Forces**	Classification of force systems, Principle of transmissibility, composition and resolution of forces. Resultant of coplanar force system (Concurrent forces, parallel forces and general system of forces). Moment of force about a point, Couples, Varignon's Theorem. Resultant of Non-Coplanar (Space Force): Concurrent force system	04	CO1
II	**Centroid**	Centroids of plane laminas: Plane lamina consisting of primitive geometrical shapes.	03	CO2
III	**Equilibrium of Force system and Friction**	**3.1 Equilibrium**: Conditions of equilibrium for concurrent forces, parallel forces and general forces, Couples. Equilibrium of rigid bodies, free body diagrams. **3.2 Equilibrium of Beams:** Types of beams, simple and compound beams, type of supports and reaction: Determination of reactions at supports for various types of loads on beams. (Excluding problems on internal hinges) **3.3 Friction:** Laws of friction. Cone of friction. angle of repose, angle of friction, equilibrium of bodies on a horizontal and inclined plane.	06	CO3
IV	**Kinematics of particle and rigid bodies**	**4.1** Motion of particle with variable acceleration. Motion along plane curved path. velocity and acceleration in terms of rectangular components, tangential and normal component of acceleration. **4.2** Introduction to general plane motion, problem based on Instantaneous center (ICR) method for general plane motion (up to 2 linkage mechanism and no relative velocity method)	05	CO4

V	**Kinetics of particle**	**5.1** Force and Acceleration: -Introduction to basic concepts, D'Alembert's Principle, concept of Inertia force, Equations of dynamic equilibrium. **5.2** Principle of linear impulse and momentum. Impact and collision: Law of conservation of momentum, Coefficient of Restitution. Direct Central Impact and Oblique Central Impact. Loss of Kinetic Energy in collision of inelastic bodies.	**05**	**CO5**
VI	**Introduction to Robot Kinematics**	Fundamental of Robot Mechanics, Degree of Freedom, D-H Parameters, robot kinematics (Forward), Homogeneous transformation (limited to 2 DOF Serial robot)	**02**	**CO6**

Text Books:

1. Engineering Mechanics by A K Tayal, Umesh Publication.
2. Engineering Mechanics by Kumar, Tata McGraw Hill
3. Engineering Mechanics by Beer &Johnston, Tata McGraw Hill

References:

1. Engineering Mechanics by R. C. Hibbeler.
2. Engineering Mechanics by F. L. Singer, Harper& Raw Publication
3. Engineering Mechanics by Macklin & Nelson, Tata McGraw Hill
4. Engineering Mechanics by Shaum Series
5. Engineering Mechanics (Statics) by Meriam and Kraige, Wiley Bools
6. Engineering Mechanics (Dynamics) by Meriam and Kraige, Wiley Bools
7. Introduction to Industrial Robotics by Ramchandran Nagrajan, Pearson publication

Online References:

Sr. No.	Website Name
3.	https://archive.nptel.ac.in/courses/112/106/112106286/
4.	https://onlinecourses.nptel.ac.in/noc21_me70/preview
3.	https://archive.nptel.ac.in/courses/112/106/112106180/

Assessment:

Internal Assessment Test (IAT) for 20 marks each:

- IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

End Semester Theory Examination:

- **Question paper format**

  - Question Paper will comprise a total of **six questions each carrying 15 marks Q.1** will be **compulsory** and should **cover the maximum contents of the syllabus**

  - **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from

Module 3 then part (b) must be from any other Module randomly selected from all the modules).

- A total of **four questions** needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
ESL101	Engineering Mechanics Lab	--	02	-	--	01	-	01

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment Test (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
ESL101	Engineering Mechanics Lab	--	--	--	--	--	25	25	50

Lab Objectives:

1. To acquaint with basic principles of centroid and its application
2. To familiarize with the concepts of force, moment, Resultant and Equilibrium of system of coplanar force.
3. To acquaint with the basic concept of friction and its application in real-life problems.
4. To understand the parameters required to quantify the Kinematics of Particle and Rigid body.
5. To understand the parameters required to quantify the Kinetics of rigid body.
6. To acquaint with the basics of Robot kinematics

Lab Outcomes:

1. Determine the equivalent force-couple system for a given system of forces (L3)
2. Demonstrate the understanding of Centroid and its significance and locate the same. (L3)
3. Illustrate the concept of force, moment and apply the same along with the concept of equilibrium in two- and three-dimensional systems with the help of FBD. (L3)
4. Calculate position, velocity and acceleration etc of particle and rigid body using principles of kinematics. (L3)
5. Analyze particles in motion using force and acceleration, work-energy and impulse-momentum principles (L4)
6. Establish the relation between robot joints and parameters (L2)

List of Experiments:

Minimum six experiments from the following list of which a minimum one should be from dynamics.

Sr No	List of Experiments	Hrs	CO mapping
01	Verification of Polygon law of coplanar forces	**01**	**LO1**
02	Verification of the Principle of Moments (Bell crank lever)	**01**	**LO3**
03	Determination of support reactions of a Simply Supported Beam.	**01**	**LO3**
04	Determination of coefficient of friction) using inclined plane	**01**	**LO3**
05	Verification of the equations of equilibrium for non-concurrent non-parallel (General)force system.	**02**	**LO3**
06	Collision of elastic bodies (Law of conservation of momentum).	**02**	**LO5**
07	Kinematics of particles. (Uniform motion of a particle, Projectile motion, motion undergravity)	**02**	**LO4**
08	Kinetics of particles. (collision of bodies)	**02**	**LO5**

Sr No	List of Assignments / Tutorials	Hrs	CO mapping
01	Resultant of Coplanar force system	**02**	**LO1**
02	Resultant of non-coplanar force system: Concurrent force system	**01**	**LO1**
03	Centroid of Composite plane Laminas	**01**	**LO2**
04	Equilibrium of System of Coplanar Forces including support reaction of beams	**02**	**LO3**
05	Equilibrium of bodies on inclined plane and problems involving ladder.	**02**	**LO3**
06	Kinematics of particles (Variable acceleration)	**02**	**LO4**
07	Kinetics of particles (D'Alembert's Principle, Impulse momentum Principle, Impact and Collisions.)	**02**	**LO5**
08	Homogeneous transformation, Direct Kinematics of robot	**02**	**LO6**

Term Work: Term Work shall consist of at least 6 practical's and 8 assignments based on the above list

Term Work Marks: 25 Marks (Total marks) = 10 Marks (Experiment) + 10 Marks (Assignments) + 5 Marks (Attendance)

Oral Exam: An Oral exam will be held based on entire syllabus.

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
ESC102	**Basic Electrical and Electronics Engineering**	**3**	**--**	**-**	**3**	**--**	**-**	**3**